EXP NO:1.a Analyze the trend of data science job postings over the last decade for the given data on the number of data science job postings each year. Use pandas for data manipulation and matplotlib/seaborn for visualization.

AIM:

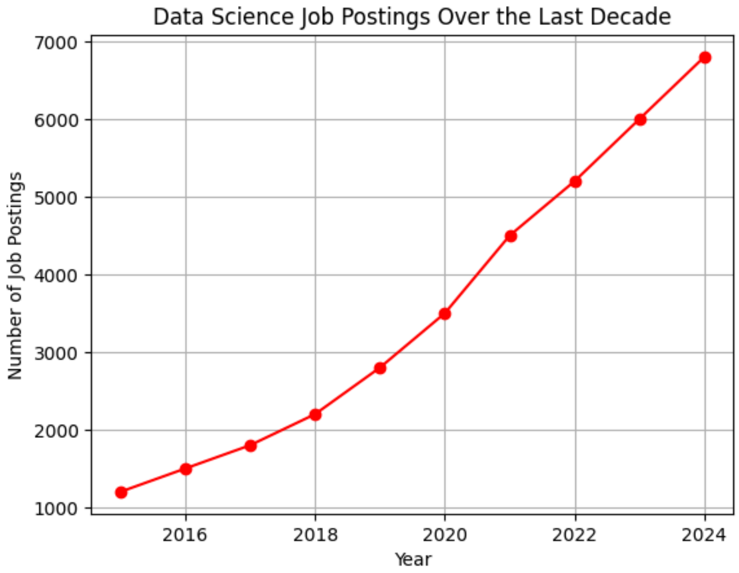
To analyze the trend of data science job postings over the last decade for the given data on the number of data science job postings each year, using pandas for data manipulation and matplotlib/seaborn for visualization.

PROGRAM:

python

```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    "Year": [2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024],
    "Job_Postings": [1200, 1500, 1800, 2200, 2800, 3500, 4500, 5200, 6000, 6800]
}

df = pd.DataFrame(data)
print(df)

plt.plot(df["Year"], df["Job_Postings"], marker='o', color="red")
plt.title("Data Science Job Postings Over the Last Decade")
plt.xlabel("Year")
plt.ylabel("Number of Job Postings")
plt.grid(True)
plt.show()
```

OUTPUT:



RESULT:

Thus the program to analyze the trend of data science job postings over the last decade was executed successfully.

EXP NO:1.b Analyze and visualize the distribution of various data science roles (Data Analyst, Data Engineer, Data Scientist, etc.) from a dataset.

AIM:

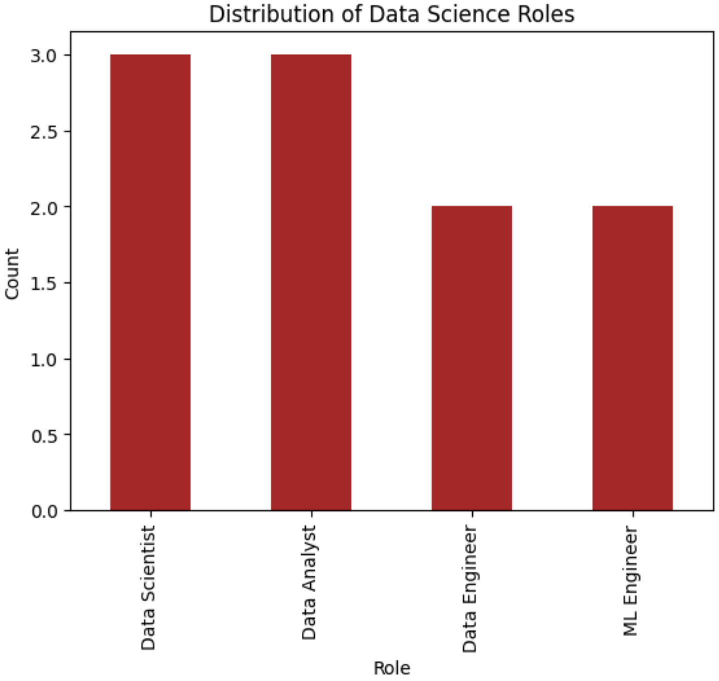
To analyze and visualize the distribution of various data science roles (Data Analyst, Data Engineer, Data Scientist, etc.) from a dataset.

PROGRAM:

python

```
import pandas as pd
import matplotlib.pyplot as plt

data = {
 "Role": ["Data Scientist", "Data Analyst", "Data Engineer",
          "ML Engineer", "Data Analyst", "Data Scientist",
          "Data Engineer", "Data Scientist", "ML Engineer",
          "Data Analyst"]
}

df = pd.DataFrame(data)
role_count = df["Role"].value_counts()
print(role_count)

role_count.plot(kind="bar", color="brown")
plt.title("Distribution of Data Science Roles")
plt.xlabel("Role")
plt.ylabel("Count")
plt.show()
```

OUTPUT:



RESULT:

Thus the program to analyze and visualize the distribution of various data science roles from a dataset was executed successfully.

EXP NO:1.c Conduct an experiment to differentiate Structured, Un-structured and Semi structured data based on data sets given.

AIM:

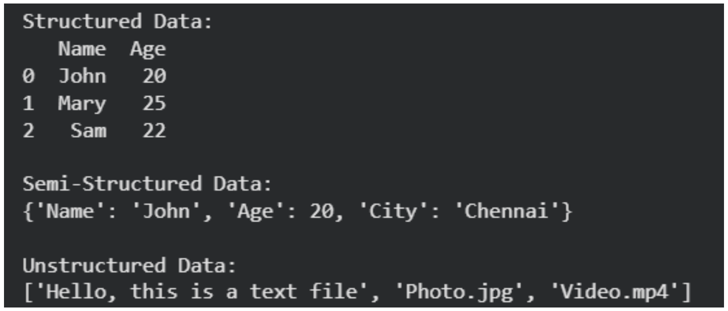
To conduct an experiment to differentiate Structured, Un-structured and Semi structured data based on data sets given.

PROGRAM:

python

```
import pandas as pd

structured_data = pd.DataFrame({
    "Name": ["John", "Mary", "Sam"],
    "Age": [20, 25, 22]
})

semi_structured_data = {
    "Name": "John",
    "Age": 20,
    "City": "Chennai"
}

unstructured_data = [
    "Hello, this is a text file",
    "Photo.jpg",
    "Video.mp4"
]

print("Structured Data:")
print(structured_data)

print("\nSemi-Structured Data:")
print(semi_structured_data)

print("\nUnstructured Data:")
print(unstructured_data)
```

OUTPUT:



```
Structured Data:
   Name  Age
0  John   20
1  Mary   25
2   Sam   22

Semi-Structured Data:
{'Name': 'John', 'Age': 20, 'City': 'Chennai'}

Unstructured Data:
['Hello, this is a text file', 'Photo.jpg', 'Video.mp4']
```

RESULT:

Thus the program to differentiate structured, semi-structured and unstructured data was executed successfully.

EXP NO:1.d Conduct an experiment to encrypt and decrypt given sensitive data. Use the cryptography library Fernet.

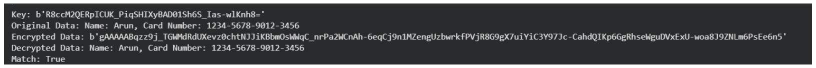
AIM:

To conduct an experiment to encrypt and decrypt given sensitive data using the cryptography library Fernet.

PROGRAM:

```
python
# Install the library if it is not already available
# !pip install cryptography

from cryptography.fernet import Fernet

# Generate a secret key
key = Fernet.generate_key()
print("Key:", key)

# Create a Fernet object using the key
cipher = Fernet(key)

# Sensitive data to be encrypted
message = "Name: Arun, Card Number: 1234-5678-9012-3456"
print("Original Data:", message)

# Encrypt the data
encrypted_data = cipher.encrypt(message.encode())
print("Encrypted Data:", encrypted_data)

# Decrypt the data
decrypted_data = cipher.decrypt(encrypted_data).decode()
print("Decrypted Data:", decrypted_data)

# Check that the decrypted data matches the original
print("Match:", message == decrypted_data)
```

OUTPUT:



```
Key: b'R8ccM2QERpICUK_PiqSHIXyBAD01Sh6S_Ias-wlKnh8='
Original Data: Name: Arun, Card Number: 1234-5678-9012-3456
Encrypted Data: b'gAAAAABqzz9j_TGWMdRdUXevz0chtNJJiKBbmOsWWqC_nrPa2WCnAh-6eqCj9n1MZengUzbwrkfPVjR8G9gX7uiYiC3Y97Jc-CahdQIKp6GgRhseWguDVxExU-woa8J9ZNLm6PsEe6n5'
Decrypted Data: Name: Arun, Card Number: 1234-5678-9012-3456
Match: True
```

(paste your output here)

RESULT:

Thus the program to encrypt and decrypt sensitive data using the cryptography library Fernet was executed successfully.